Milestone 2 — In-house FSDT Solver: Theory, Implementation, Verification & Wing Results (Step 2)

Deliverable of this step: *"Validated FSDT MATLAB Solver"* **Scripts:**

- `validation/run_all_validation.m`: 9 benchmark groups, **42/42 required checks pass**.
- `milestone2/run_milestone2.m`: wing analyses (static, modal, dynamic, damage).

Logs: `results/data/validation_summary.txt`, `results/data/M2_summary.txt`

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1. Why an in-house FSDT solver?

Need of the digital twin	What the solver gives
Thousands of DOE runs (Milestone 3)	~0.25 s per complete static + modal analysis, scriptable, no licence
Physics residuals for the PINN loss (Milestone 5)	We own every equation: K, M, the B-operators and the shear stiffness As can be exported
Damage of any shape	Element-wise section properties (continuum damage)
Correct sandwich physics	Energy-equivalent shear correction + transverse-shear stress recovery
Real-time surrogate (Milestone 6)	The same K, M feed reduced-order models and Kalman filters

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2. First-order Shear Deformation Theory (Reissner–Mindlin)

2.1 Kinematics

$$u(x,y,z,t) = u_0 + z\psi_x, \qquad v(x,y,z,t) = v_0 + z\psi_y, \qquad w(x,y,z,t) = w_0.$$

There are five unknown fields: u_o, v_o, w_o, ψ_x, ψ_Y.

- **CLT/Kirchhoff** forces $\psi_x = -\partial w/\partial x$, which means the normal stays perpendicular and the shear strain is zero.
- **FSDT** lets the normal rotate independently, so transverse shear strain exists.

2.2 Strains

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^0 + z\boldsymbol{\kappa}: \quad \boldsymbol{\varepsilon}^0 = \begin{Bmatrix} u_{0,x} \\ v_{0,y} \\ u_{0,y} + v_{0,x} \end{Bmatrix}, \ \boldsymbol{\kappa} = \begin{Bmatrix} \psi_{x,x} \\ \psi_{y,y} \\ \psi_{x,y} + \psi_{y,x} \end{Bmatrix}, \qquad \boldsymbol{\gamma} = \begin{Bmatrix} \gamma_{xz} \\ \gamma_{yz} \end{Bmatrix} = \begin{Bmatrix} w_{0,x} + \psi_x \\ w_{0,y} + \psi_y \end{Bmatrix}.$$

2.3 Stress resultants and constitutive law

$$(\mathbf{N}, \mathbf{M}) = \int_{-h/2}^{h/2} \boldsymbol{\sigma}\,(1, z)\,dz, \qquad \mathbf{Q} = \int \boldsymbol{\tau}\,dz$$

$$\begin{Bmatrix} \mathbf{N} \\ \mathbf{M} \end{Bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{D} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\varepsilon}^0 \\ \boldsymbol{\kappa} \end{Bmatrix}, \qquad \mathbf{Q} = \mathbf{A}_s \boldsymbol{\gamma}, \quad A_{s,ij} = k_{ij} \sum_k \bar{Q}^{(k)}_{s,ij} (z_k - z_{k-1})$$

The shear correction factors k come from the energy method of Milestone 1.

2.4 Governing equations (Hamilton's principle)

$$\delta \int_{t_1}^{t_2} (T - U + W)\,dt = 0$$

This gives the five FSDT equations of motion:

$$\begin{aligned} N_{x,x} + N_{xy,y} &= I_0 \ddot{u}_0 + I_1 \ddot{\psi}_x, & N_{xy,x} + N_{y,y} &= I_0 \ddot{v}_0 + I_1 \ddot{\psi}_y, \\ Q_{x,x} + Q_{y,y} + q &= I_0 \ddot{w}_0, & & \\ M_{x,x} + M_{xy,y} - Q_x &= I_1 \ddot{u}_0 + I_2 \ddot{\psi}_x, & M_{xy,x} + M_{y,y} - Q_y &= I_1 \ddot{v}_0 + I_2 \ddot{\psi}_y. \end{aligned}$$

These five equations are the physics the PINN will enforce in Milestone 5.

2.5 Weak form (principle of virtual work)

$$\int_A (\delta\boldsymbol{\varepsilon}^{0T} \mathbf{N} + \delta\boldsymbol{\kappa}^T \mathbf{M} + \delta\boldsymbol{\gamma}^T \mathbf{Q}) dA + \int_A \delta\mathbf{u}^T \mathbf{m}\,\ddot{\mathbf{u}}\,dA = \int_A \delta w\, q\, dA,$$

$$\mathbf{m} = \begin{bmatrix} I_0 & 0 & 0 & I_1 & 0 \\ 0 & I_0 & 0 & 0 & I_1 \\ 0 & 0 & I_0 & 0 & 0 \\ I_1 & 0 & 0 & I_2 & 0 \\ 0 & I_1 & 0 & 0 & I_2 \end{bmatrix}.$$

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3. Finite-element discretisation

3.1 Element

- 4-node isoparametric quadrilateral (`src/element/elem_fsdt_q4.m`).
- 5 DOF per node, [u v w ψ_x ψ_y], so **20 DOF per element**.
- Bilinear shape functions $N_a = \frac{1}{4}(1 + \xi_a \xi)(1 + \eta_a \eta)$.
- Jacobian $\mathbf{J} = \begin{bmatrix} x_{,\xi} & y_{,\xi} \\ x_{,\eta} & y_{,\eta} \end{bmatrix}$ and $\partial N/\partial(x,y) = \mathbf{J}^{-1}\,\partial N/\partial(\xi,\eta)$.

Generalised strains are written ε^0 = B_m d, κ = B_b d, γ = B_s d, so that

$$\mathbf{K}_e = \int_{A_e} \begin{bmatrix} \mathbf{B}_m \\ \mathbf{B}_b \end{bmatrix}^T \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{B}_m \\ \mathbf{B}_b \end{bmatrix} dA + \mathbf{K}_s, \qquad \mathbf{M}_e = \int_{A_e} \mathbf{N}^T \mathbf{m}\,\mathbf{N}\,dA \text{ (consistent, incl. rotary inertia)}$$

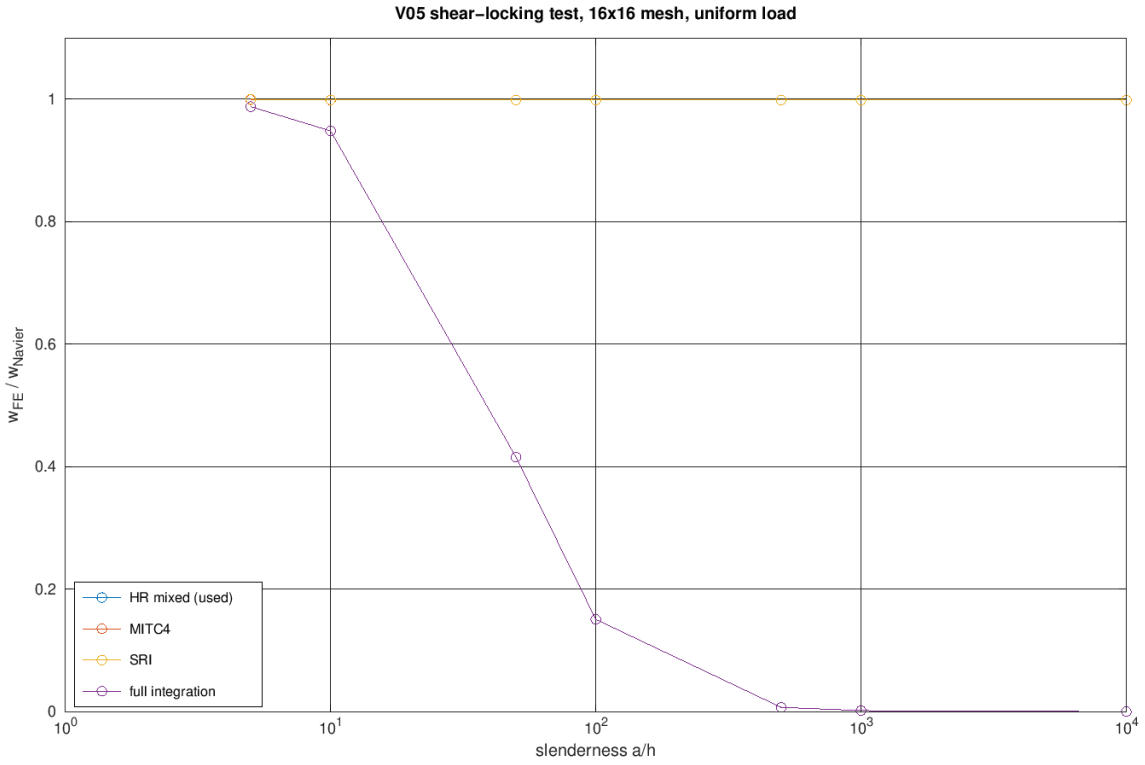
Both are integrated with 2 × 2 Gauss quadrature, which is exact for parallelograms.

3.2 The central difficulty: shear locking

As h → 0, FSDT must recover Kirchhoff theory, which requires $\gamma = w_{,x} + \psi_x \to 0$.

- With bilinear w and ψ, the constraint $w_{,x} + \psi_x = 0$ at every Gauss point is too strong.
- The element can then only satisfy it by not bending at all. It **locks**: deflection goes to zero.

Validation V05: the fully integrated element gives 15 % of the correct deflection at a/h = 100 and 0.002 % at a/h = 10^4.



Shear locking

Remedy	Locking-free	Rigid-body modes	Distorted patch test	Comment
Full integration	×	6	—	reference only
Selective reduced integration (1-pt shear)	✓	**8 (2 spurious hourglass modes)**	—	dangerous with weak supports
MITC4 (assumed strain)	✓	6	✓	Bathe–Dvorkin 1985
Hellinger–Reissner mixed (used)	✓	**6**	**✓ ($1.8\cdot10^{-11}$)**	gives shear forces directly

3.3 Hellinger–Reissner (HR) mixed formulation — derivation

The HR principle treats **displacements and stresses as independent fields**. For the transverse-shear part:

$$\Pi_{HR}(\mathbf{u},\mathbf{Q}) = \int_A (\mathbf{Q}^T\bar{\boldsymbol{\gamma}}(\mathbf{u}) - \tfrac{1}{2}\mathbf{Q}^T\mathbf{A}_s^{-1}\mathbf{Q})dA.$$

- Stationarity with respect to Q gives the constitutive law **weakly**: ∫ δQᵀ(γ̄ − A_s⁻¹Q) dA = 0.
- Stationarity with respect to u gives equilibrium.

Ingredient 1 — assumed covariant shear strain (tying). On each element edge, the displacement-based covariant strain at the edge **mid-point** is exact for any quadratic w and linear ψ. The strain field is therefore built by interpolating between those four points (A, C on the η = ∓1 edges; D, B on the ξ = ∓1 edges):

$$\bar{\gamma}_\xi = \tfrac{1-\eta}{2}\gamma_\xi^A + \tfrac{1+\eta}{2}\gamma_\xi^C, \qquad \bar{\gamma}_\eta = \tfrac{1-\xi}{2}\gamma_\eta^D + \tfrac{1+\xi}{2}\gamma_\eta^B, \qquad \gamma_\xi = x_{,\xi}\gamma_{xz} + y_{,\xi}\gamma_{yz}.$$

Ingredient 2 — assumed shear-force field, Piola-mapped.

$$\hat{\mathbf{Q}} = \begin{bmatrix} 1 & \eta & 0 & 0 \\ 0 & 0 & 1 & \xi \end{bmatrix}\boldsymbol{\beta} = \hat{\mathbf{P}}\boldsymbol{\beta}, \qquad \mathbf{Q} = \frac{\mathbf{J}^T\hat{\mathbf{Q}}}{\det\mathbf{J}} = \mathbf{P}\boldsymbol{\beta}.$$

The Piola map makes the coupling term exact in the parent square: $\mathbf{Q}^T\bar{\boldsymbol{\gamma}}\,dA = \hat{\mathbf{Q}}^T\bar{\boldsymbol{\gamma}}_{cov}\,d\xi\,d\eta$.

Element condensation. Substituting the two fields:

$$\Pi_e = \boldsymbol{\beta}^T\mathbf{G}\,\mathbf{d} - \tfrac{1}{2}\boldsymbol{\beta}^T\mathbf{H}\boldsymbol{\beta}, \qquad \mathbf{H} = \int_{A_e}\mathbf{P}^T\mathbf{A}_s^{-1}\mathbf{P}\,dA, \qquad \mathbf{G} = \int_{-1}^{1}\int_{-1}^{1}\hat{\mathbf{P}}^T\bar{\mathbf{B}}_{cov}\,d\xi\,d\eta$$

Setting ∂Π_e/∂β = 0 gives β = H⁻¹G d. Then

$$\mathbf{K}_s = \mathbf{G}^T\mathbf{H}^{-1}\mathbf{G}, \qquad \mathbf{Q}_{centre} = \mathbf{P}(0,0)\,\mathbf{H}^{-1}\mathbf{G}\,\mathbf{d}.$$

Properties, all verified numerically:

- Exactly 6 zero-energy modes (V01).
- Passes the MacNeal–Harder distorted patch test for constant membrane strain and constant curvature, at a/h = 12 and 240, with interior error ≤ $1.8\cdot10^{-11}$ (V02).
- No locking from a/h = 5 to 10^4 (V05).
- Q is obtained from β with no differentiation of displacements, which is more accurate for the interlaminar stress recovery.

> **Note on history.** The first implementation mapped the assumed field with the centre Jacobian and used raw displacement strains. It was locking-free and correct on rectangles, but it **failed the distorted patch test (8–14 % error)**. Adding the tied covariant strains (ingredient 1) and the Piola map (ingredient 2) fixed it. This is why patch tests are part of the suite.

3.4 Assembly and boundary conditions

- **Sparse triplet assembly** (`src/assembly/assemble_system.m`). An element-matrix cache keyed by (section, shape) means a uniform 400-element wing computes **one** element matrix. This matters for the DOE.
- **Dirichlet BCs** are applied by partitioning into free and fixed DOFs (`apply_bc.m`):
  - clamped root (wing);
  - SS-1 simply supported for the Navier benchmarks: x = 0, a: v = w = ψ_γ = 0; y = 0, b: u = w = ψ_x = 0.

3.5 Solvers

Analysis	Equation	Implementation
Static	K_ff d_f = F_f	sparse direct; reactions R = Kd − F
Modal	$(K - \omega^2 M)\varphi = 0$	`eig` (≤ 1500 DOF) or shift-invert `eigs`; mass-normalised $\varphi^T M \varphi = 1$; **sign fixed** (largest w positive) so ML features are deterministic
Transient	$M\ddot{d} + C\dot{d} + K d = F(t)$	**Newmark-β**, γ = ½, β = ¼ (average acceleration): unconditionally stable, 2nd-order accurate, no numerical damping; K_eff factorised once with sparse Cholesky
Damping	$C = \alpha M + \beta K$	Rayleigh, $\zeta(\omega) = \alpha/2\omega + \beta\omega/2$, fitted to ζ = 2 % at modes 1 and 5

Newmark update, with $a_0 = 1/(\beta \Delta t^2)$, $a_1 = \gamma/(\beta \Delta t)$, ...:

$$(\mathbf{K} + a_0\mathbf{M} + a_1\mathbf{C})\,\mathbf{d}_{n+1} = \mathbf{F}_{n+1} + \mathbf{M}(a_0\mathbf{d}_n + a_2\dot{\mathbf{d}}_n + a_3\ddot{\mathbf{d}}_n) + \mathbf{C}(a_1\mathbf{d}_n + a_4\dot{\mathbf{d}}_n + a_5\ddot{\mathbf{d}}_n)$$

3.6 Stress recovery and failure (`src/post/recover_stresses.m`)

1. **In-plane ply stresses** at the bottom and top of each ply: $\sigma_{xy} = \bar{Q}_k(\varepsilon^o + z\kappa)$, then rotated to material axes, $\sigma_{12} = Q_k T_\varepsilon \varepsilon$.
2. **Transverse shear through the thickness** from the equilibrium profile of Milestone 1: $\tau_{xz}(z) = Q_x f_x(z)$, with $\int f\,dz = 1$.
   - FSDT alone gives a constant, physically wrong τ.
   - The profile gives correct **interlaminar shear at ply interfaces**, the delamination driver, and correct **core shear**, the debond driver.
3. **Failure indices**, all scaled so that 1 = failure:

Criterion	Formula	Use
Tsai–Wu (as 1/R)	$F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau^2 + 2F_{12}\sigma_1\sigma_2 = 1$ solved for the load multiplier R; $F_{12} = -\frac{1}{2}\sqrt{(F_{11}F_{22})}$	interactive ply failure
Hashin (1980)	fibre tension/compression, matrix tension/compression	identifies the failure **mode** (a damage label for the AI)
Max stress	$\max(\sigma_1/X, \sigma_2/Y, \tau/S)$	sanity check
Core shear	max τ_core / S_core	core shear failure
Interlaminar	$\lVert\tau_{interface}\rVert / S_{23}$	delamination onset

3.7 Damage model (`src/damage/apply_damage.m`)

Continuum damage mechanics: property → (1 − D) · property inside an elliptical zone, with a uniform or Gaussian profile. Damaged elements get their own section, quantised into 20 severity levels so they can share cached element matrices.

Type	Degraded properties	Physical mechanism
`skin_matrix`	E_2, G_{12}, G_{23} of face plies	matrix micro-cracking
`skin_fiber`	E_1, E_2, G_{12} of face plies	fibre breakage (impact)
`debond`	core G_{13}, G_{23}	face/core disbond, loss of shear transfer
`core_crush`	all core moduli	impact core crushing
`delam`	G_{13}, G_{23} at the mid-face interface	inter-ply delamination (equivalent single layer)

3.8 Analytical reference — Navier solution (`src/analytical/navier_fsdt.m`)

For SS-1 cross-ply plates, the double-Fourier series

$$u = U\cos\alpha x\sin\beta y,\ v = V\sin\alpha x\cos\beta y,\ w = W\sin\alpha x\sin\beta y,\ \psi_x = X\cos\alpha x\sin\beta y,\ \psi_y = Y\sin\alpha x\cos\beta y,$$

with $\alpha = m\pi/a$ and $\beta = n\pi/b$, satisfies every boundary condition exactly. Substituting into the five equations of Section 2.4 gives a 5 × 5 algebraic system per (m, n):

$$([\mathbf{S}] - \omega^2[\mathbf{M}_m])\{U\,V\,W\,X\,Y\}^T = \{0\,0\,Q_{mn}\,0\,0\}^T,$$

- $S_{33} = A_{55}\alpha^2 + A_{44}\beta^2$, $S_{44} = D_{11}\alpha^2 + D_{66}\beta^2 + A_{55}$, …; the full list is in the code header.
- Load coefficients: Q_mn = $16q_0/(\pi^2 mn)$ for a uniform load, q_0 for a sinusoidal load.

This gives an **exact** FSDT reference for the FE solver.

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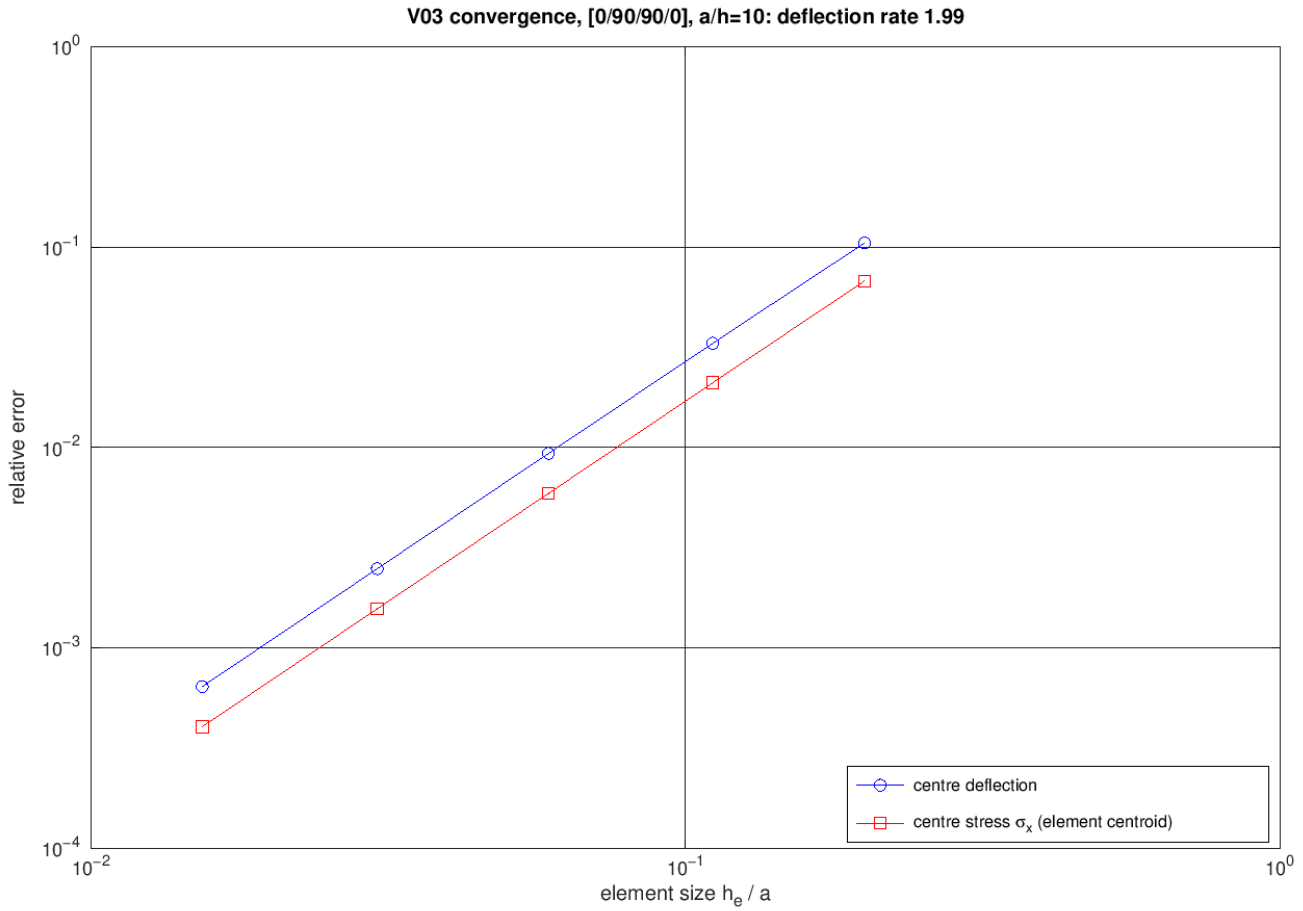
4. Verification & validation — all required checks passed (42 original + V10, V11)

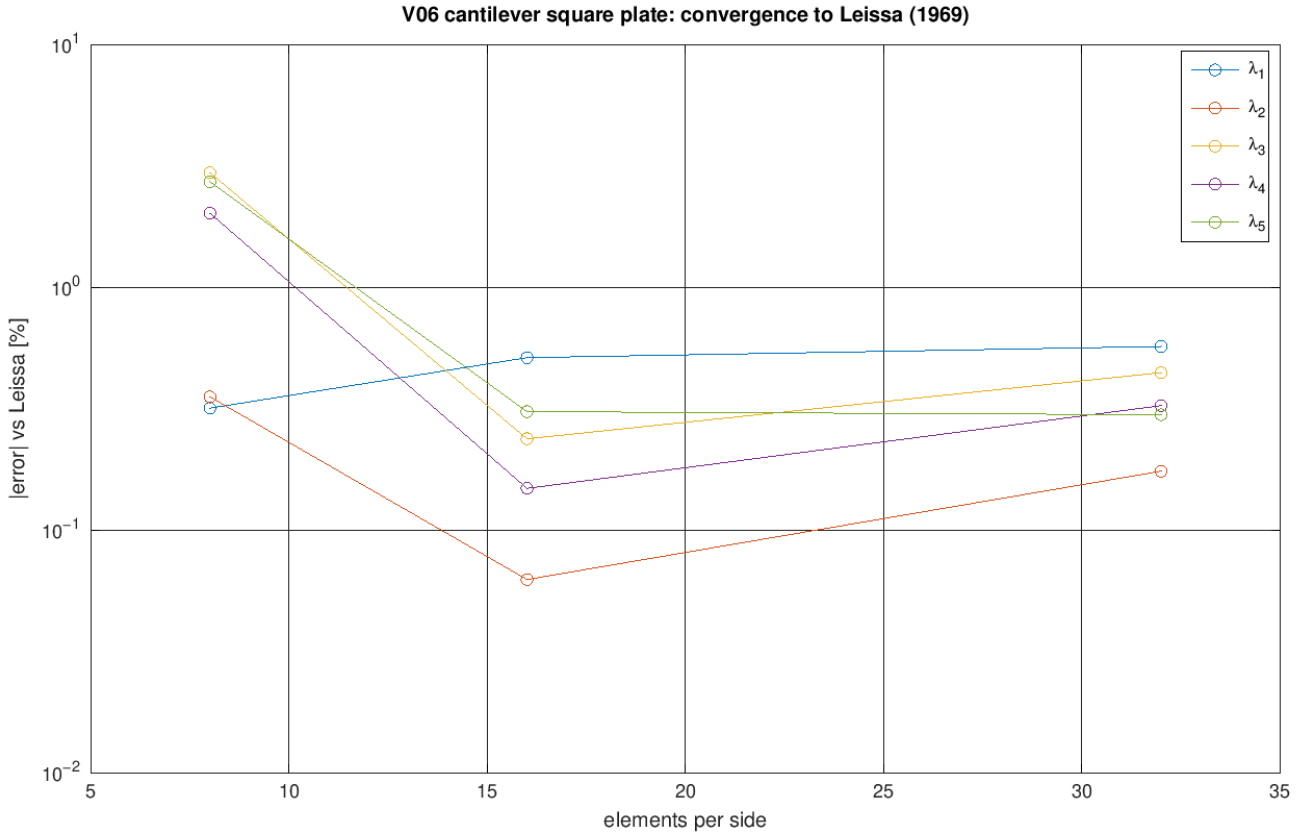
Run: `setup_paths; run_all_validation` (about 30 s).

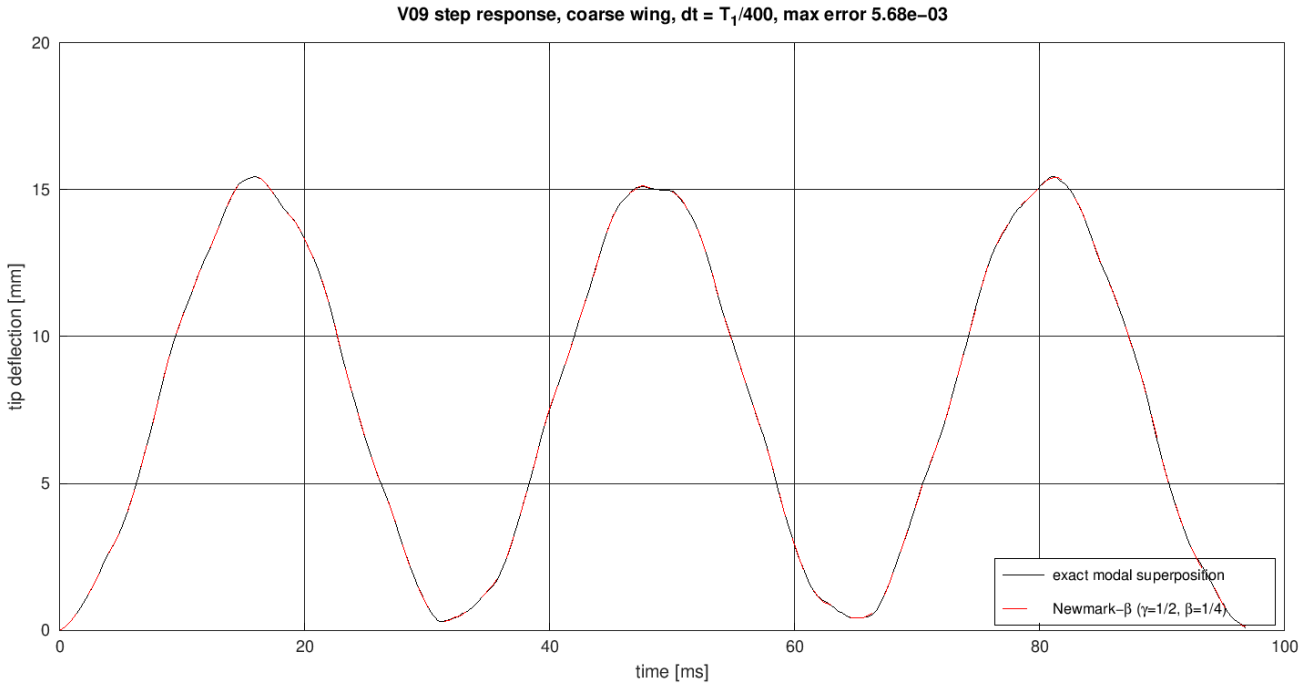
#	Benchmark	What it proves	Result
V01	Single distorted element eigenvalues	no spurious mechanisms	HR: 6 zero modes ✓ (SRI: 8 ×)
V02	MacNeal–Harder patch test, thin and thick	consistency → convergence	error ≤ $1.8\cdot10^{-11}$ ✓
V03	[0/90/90/0] SS plate, sinusoidal load, a/h = 4…100	laminate + shear + stresses	$\bar{w}$ within 0.13 %, $\bar{\sigma}_x$ within 0.3 % of Navier; **Navier = Reddy (2004) to 5 digits** (0.6627, 1.7100, 0.4912, 0.4337); convergence rate 1.99 (theory: 2)
V04	Free vibration, [0/90/90/0] and **unsymmetric [0/90]**	mass matrix, rotary inertia, B-coupling	first 6 flexural $\bar{\omega}$ within 0.76 %
V05	Shear-locking sweep, a/h = 5…10^4	locking-free	HR ≥ 0.9987 of exact at every a/h ✓

#	Benchmark	What it proves	Result
V06	Cantilever square plate (Leissa 1969)	clamped–free BCs, as on the wing	$\lambda_{1\ldots5}$ within 0.6 %
V07	Cantilever strip vs Timoshenko beam, L/h = 5 and 100	shear deflection	error 0.015 %
V08	k = 5/6 recovery; sandwich vs Allen theory	shear correction	k exact to 10^{-16}; sandwich within 0.2 % (k = 5/6 would be 29 % off)
V09	Newmark: energy, step response, DAF	transient solver	energy drift $2{\cdot}10^{-12}$; error vs exact modal solution 0.57 %; DAF 1.979 (theory 2)
V10	**Wing modes vs independent 1-D theories** (added in review)	correctness of the wing mode shapes	bending 1B–5B within 0.5–0.9 % of a Timoshenko beam, **MAC = 1.0000**; 1T within 0.08 % of Reissner strip torsion, MAC = 1.0000
V11	8-mode exact modal superposition vs full Newmark (added for Milestone 3)	fast solver used for every DOE sample	deflection 0.03 %, acceleration 0.04 %

The full table with every value, reference, relative error and tolerance is in `results/data/validation_table.md`.


V03 convergence, [0/90/90/0], a/h=10: deflection rate 1.99
relative error
element size h_e / a
centre deflection
centre stress σ_x (element centroid)


V06 cantilever square plate: convergence to Leissa (1969)
|error| vs Leissa [%]
elements per side
λ_1
λ_2
λ_3
λ_4
λ_5



Two benchmark lessons worth writing in the thesis:

- **V04 in-plane modes.** The FE spectrum contains in-plane shear modes (m = 0, $\bar{\omega} = \pi\sqrt{(G_{12}/E_2)}\cdot a/h \approx 24.3$) that the flexural Navier series excludes. They must be filtered out by their kinetic-energy share before the comparison.
- **V03 stress location.** For stresses compare at element centroids: use an odd mesh so a centroid sits at the plate centre. For deflection, use an even mesh so a node sits there.

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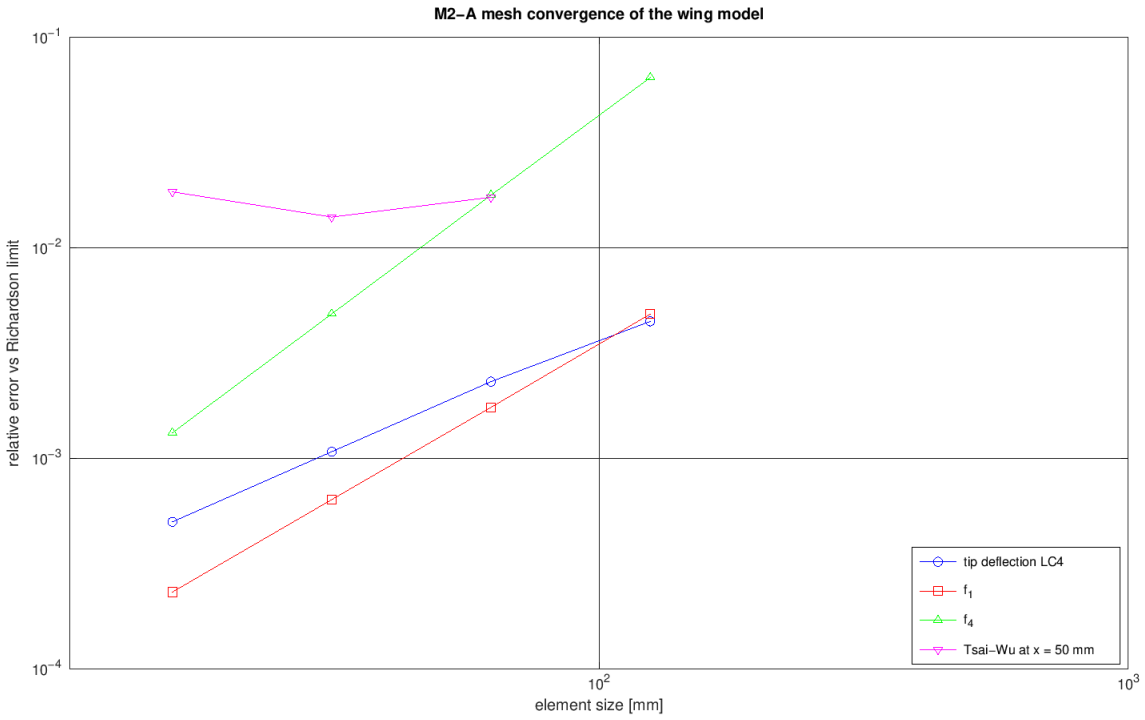
5. Results for the baseline UAV wing

5.1 Mesh convergence (Richardson extrapolation)

Meshes 8×2 → 64×16 with a constant element aspect ratio and refinement ratio 2:

$$p = \frac{\ln\left[(q_2 - q_1)/(q_3 - q_2)\right]}{\ln 2}, \qquad q_\infty = q_3 + \frac{q_3 - q_2}{2^p - 1}.$$

Quantity	Observed order p	Converged value	**Baseline 40×10 error**
Tip deflection, LC4	1.11	5.897 mm	0.08 %
$f_1 \dots f_6$	1.46 – 1.99	30.92, 156.9, 162.5, 197.5, 377.8, 457.6 Hz	0.05 – 0.32 %
Tsai–Wu at x = 50 mm	—	0.0592 – 0.0595 on all meshes	< 0.5 % spread
Peak Tsai–Wu (root TE corner)	**−0.51 (diverges)**	—	—



Convergence

> **Stress singularity at the clamped corner.** Where a clamped edge meets a free edge, plate theory predicts stresses that grow without bound as r → 0 (Williams-type corner singularity). The peak Tsai–Wu therefore *keeps growing with refinement*: 0.057 → 0.062 → 0.069 → 0.079. This is physics, not a bug. Design values are read at **x = 50 mm (≈ 2h from the root)**, where they converge. In the real aircraft, the root fitting spreads this load.

5.2 Static analysis — all load cases (40×10 mesh, 2255 DOF)

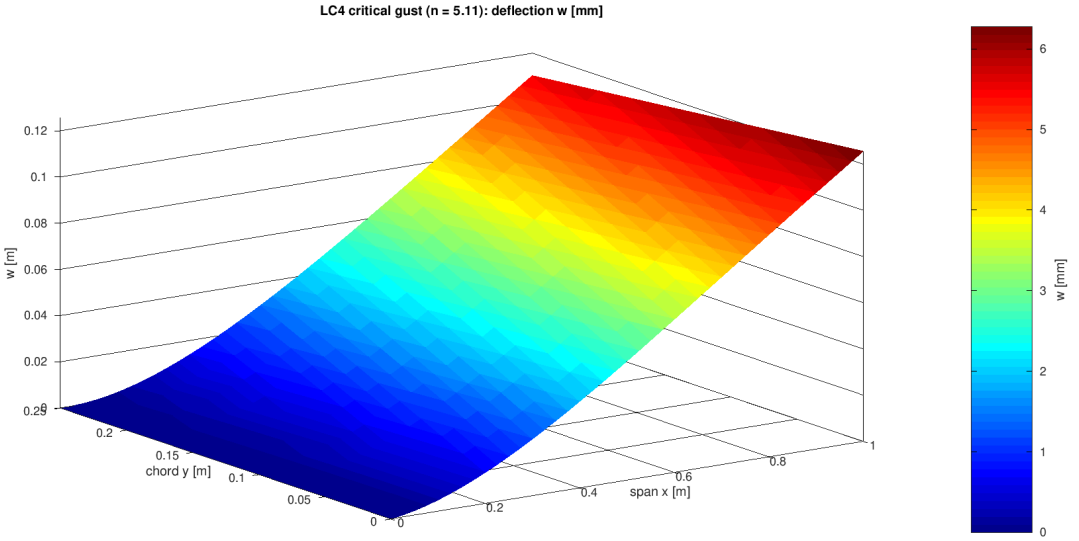
Case	n	w_LE [mm]	w_mid [mm]	w_TE [mm]	Tip twist [°]	ε_x max [με]	Tsai–Wu	Hashin	Core shear	ILSS
LC1 cruise	1.00	1.227	1.153	1.078	+0.034	55	0.014	0.014	0.008	0.008
LC2 manoeuvre +	3.80	4.664	4.380	4.098	+0.130	209	0.054	0.052	0.032	0.032
LC3 manoeuvre −	−1.50	−1.841	−1.729	−1.618	−0.051	82	0.021	0.020	0.013	0.012
LC4 critical gust	**5.11**	**6.275**	**5.892**	**5.513**	**+0.175**	**281**	0.072	0.069	0.043	0.042
LC5 ultimate	7.67	9.413	8.839	8.270	+0.262	421	0.108	0.104	0.064	0.064
LC6 tip test 50 N	—	7.789	7.896	7.780	+0.002	265	0.067	0.065	0.062	0.061

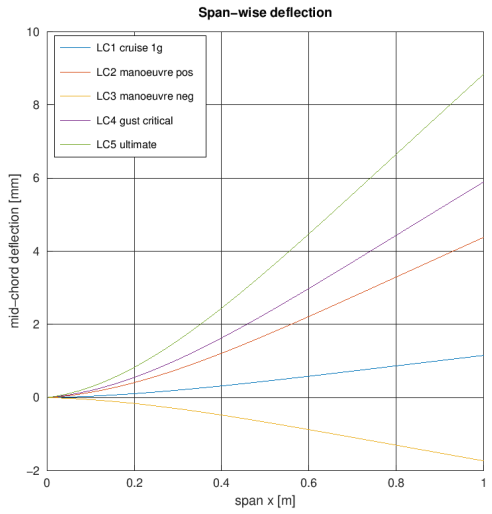
- **Global equilibrium:** $|\Sigma R_z + \Sigma F_z| \,/\, |\Sigma F_z| = 1.5 \cdot 10^{-12}$.

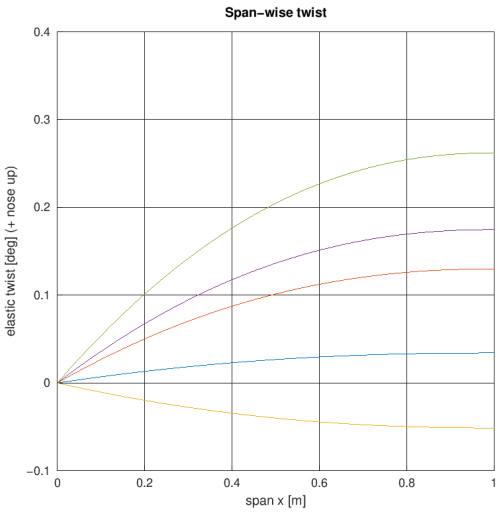
- **Nose-up twist:** the LE deflects more than the TE because lift acts at c/4, ahead of the elastic axis. This confirms the physics of Milestone 1, Section 9.2.
- **Ultimate strength (LC5):**

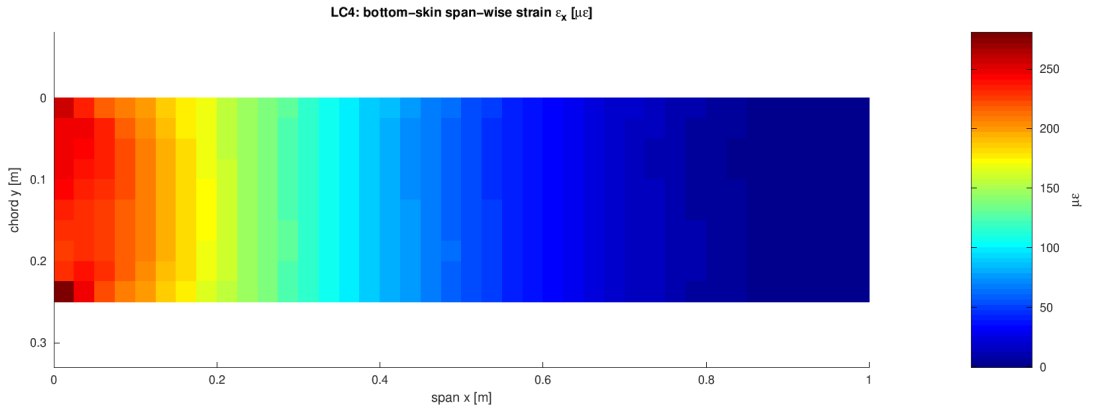
Check	Index	Reserve factor RF
Tsai–Wu peak (ply 4, 90°, root TE corner)	0.108	9.3
Tsai–Wu converged design value (x = 50 mm)	0.089	**11.2**
Hashin peak — matrix tension	0.104	9.6
Core shear	0.064	15.6
Interlaminar shear	0.064	15.8

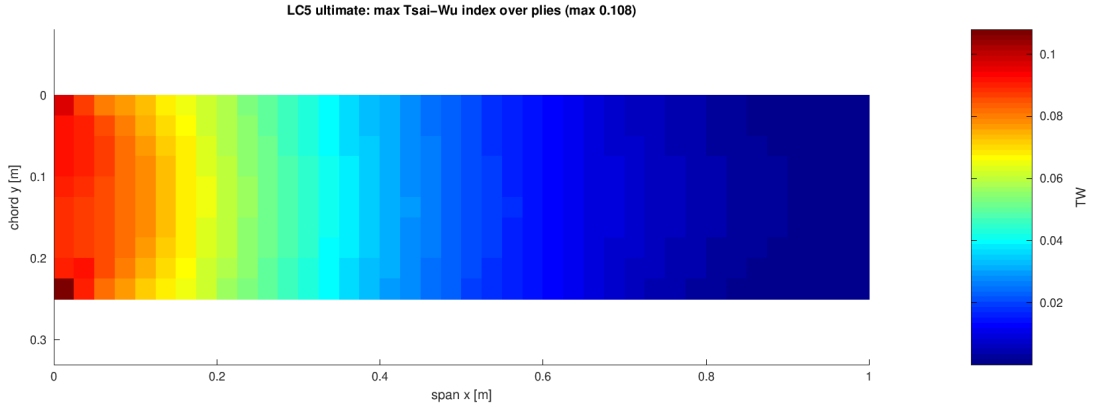
The wing is **stiffness / minimum-gauge driven**, not strength driven. That is typical for small UAVs, and it is good news for SHM: damage first shows up as a change in *stiffness* (frequencies, strains) long before it threatens *strength*.

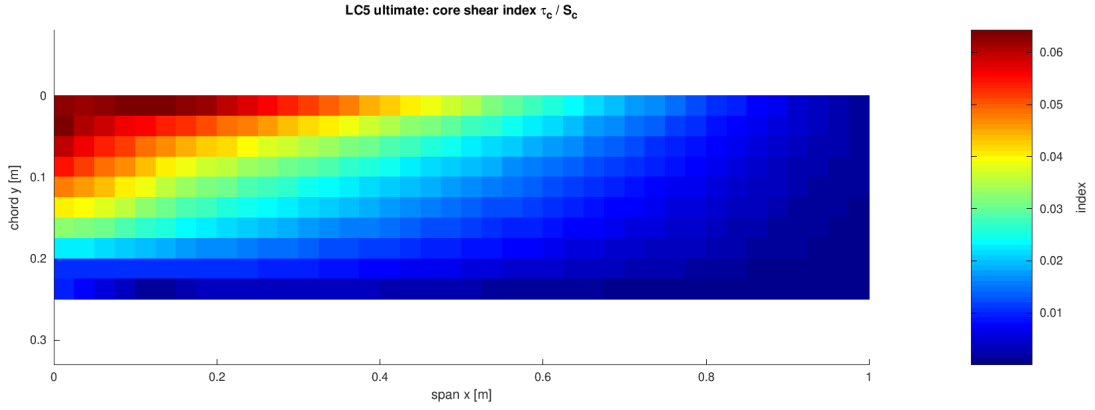


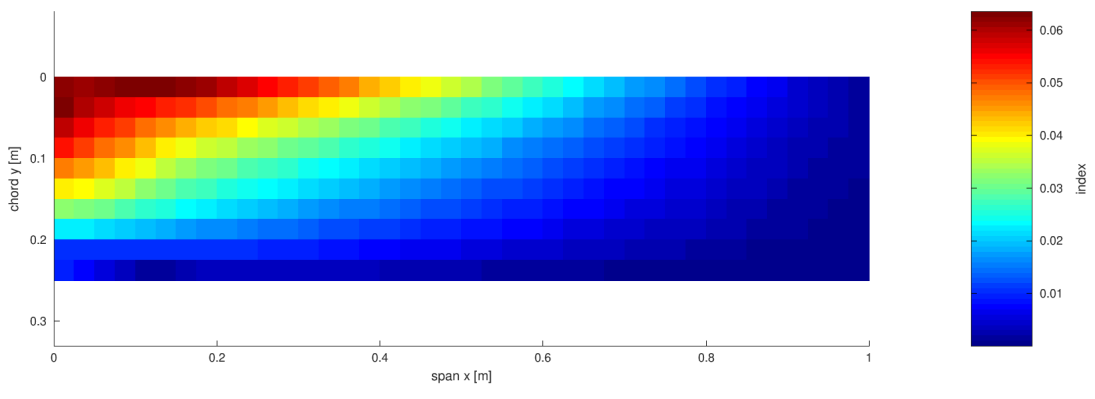












5.3 How much does transverse shear matter for this wing?

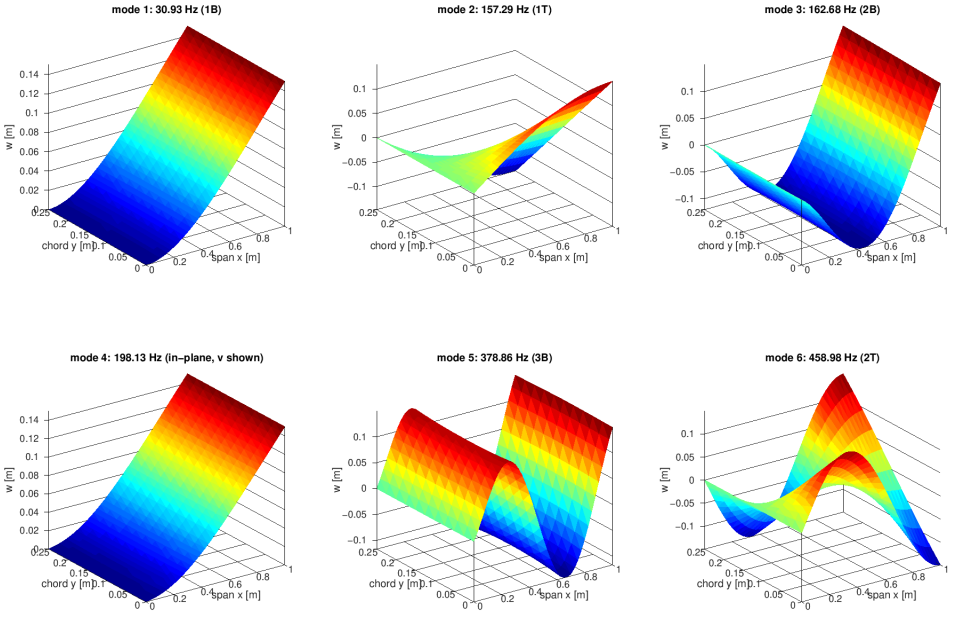
Model	Tip deflection, LC4	f_1
FSDT, energy k (used)	5.892 mm (100 %)	30.93 Hz (100 %)
FSDT, k = 5/6	5.529 mm (93.8 %)	32.03 Hz (103.5 %)
Shear-rigid (≈ CLT)	5.424 mm (92.1 %)	32.35 Hz (104.6 %)

- Core shear contributes **8 % of the deflection** and **4.6 % of f_1**.
- The damage-induced frequency shifts of Section 5.6 are 0.1 – 1.4 %. A CLT model's error on f_1 (4.6 %) would be 3 to more than 100 times larger than the signal the AI must detect, so **FSDT is mandatory for this SHM problem**.

5.4 Modal analysis

Mode	f [Hz]	Type	Comment
1	30.93	1st bending (1B)	Euler–Bernoulli beam with EI = D_{11}c: $0.56\sqrt{(EI/mL^4)}$ = 33.4 Hz; minus the 4.6 % shear effect (Section 5.3) gives ≈ 31.9 Hz ✓
2	157.29	1st torsion (1T)	
3	162.68	2nd bending (2B)	f_3/f_1 = 5.26, close to beam theory 6.27 reduced by shear and plate effects
4	198.13	in-plane chord-wise bending (lead–lag)	
5	378.86	3B	
6	458.98	2T	
7–10	618.1, 729.1, 866.2, 967.1	4B, 3T, 5B, 4T	

- **Orthogonality:** max $|\Phi^T M \Phi - I|$ = $1.8 \cdot 10^{-15}$ and max $|\Phi^T K \Phi - \Omega^2| / \omega^2_max$ = $1.3 \cdot 10^{-13}$.
- Modes are classified automatically by Legendre decomposition of w along the chord (`src/post/classify_mode.m`). These labels become DOE outputs.



Modes

Are the mode shapes correct? An independent check (V10). Orthogonality only proves the eigen-solution is self-consistent. To check the *physics*, the wing modes were compared with 1-D theories that share no code with the plate solver.

Bending is compared with a Timoshenko cantilever beam (own FE, 200 elements). Its properties come from the laminate: EI = c/d_{11} with d = D^{-1} (free edges allow anticlastic curvature), κGA = c·A_s,11, ρA = l_0c, ρI = l_2c.

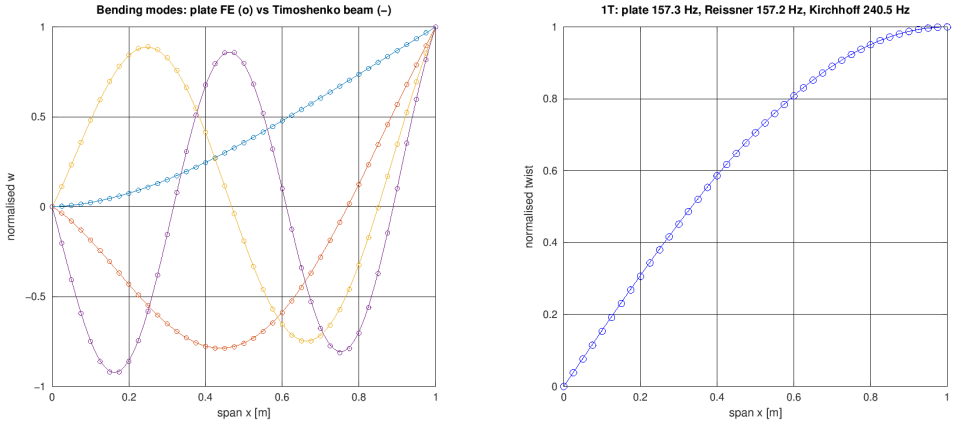
Mode	Plate FE [Hz]	Timoshenko beam [Hz]	Difference	MAC (chord-averaged w)
1B	30.931	30.733	0.65 %	1.0000
2B	162.676	161.917	0.47 %	1.0000
3B	378.864	377.095	0.47 %	1.0000
4B	618.110	614.286	0.62 %	1.0000
5B	866.213	858.898	0.85 %	1.0000

Torsion is compared with Reissner's shear-deformable plate-strip torsion:

$$GJ = 4D_{66}\,c\left[1 - \frac{\tanh(\lambda c/2)}{\lambda c/2}\right], \quad \lambda = \sqrt{A_{s,11}/D_{66}}, \quad f_{T1} = \frac{1}{4L}\sqrt{\frac{GJ}{I_0 c^3/12 + I_2 c}}$$

This gives **157.17 Hz vs 157.29 Hz from the plate (0.08 %)**, and the twist shape matches sin(πx/2L) with MAC = 1.0000.

Kirchhoff (no-shear) torsion would give 240.5 Hz, 53 % too high, because the soft core cuts the torsional stiffness to 43 %. This is one more proof that FSDT is necessary. The plate is slightly stiffer than the beam (0.5–0.9 %) because the clamped root also prevents anticlastic curvature locally, which the beam cannot represent.

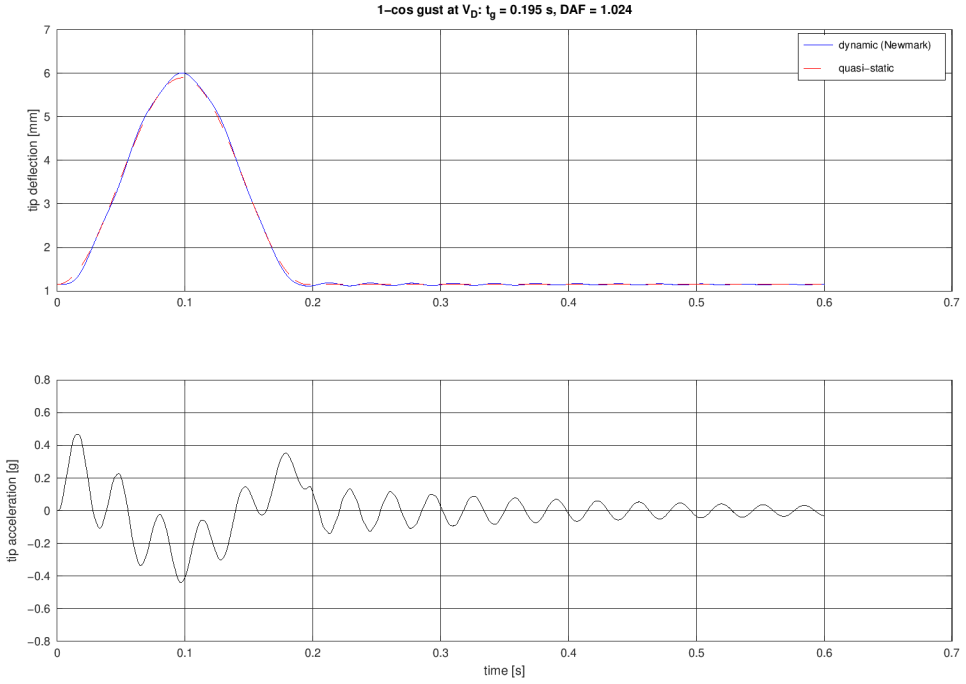


Modes vs beam

5.5 Dynamic gust response

Design gust: 1-cosine at V_D, gradient H = 12.5 $\bar{c}$ = 3.125 m, duration t_g = 2H/V = 0.195 s, so t_g / T_1 = 6.0.

- Analysis: Newmark, Δt = 0.25 ms (≈ 130 steps per T_1), ζ = 2 % Rayleigh (α = 7.19 s^{-1}, β = $1.55 \cdot 10^{-5}$ s).
- Peak tip deflection: **6.008 mm**.
- **DAF = 1.024**, so the design gust is quasi-static.
- Peak tip acceleration: 0.47 g. This is the signal amplitude an accelerometer would see (input for Milestone 6).

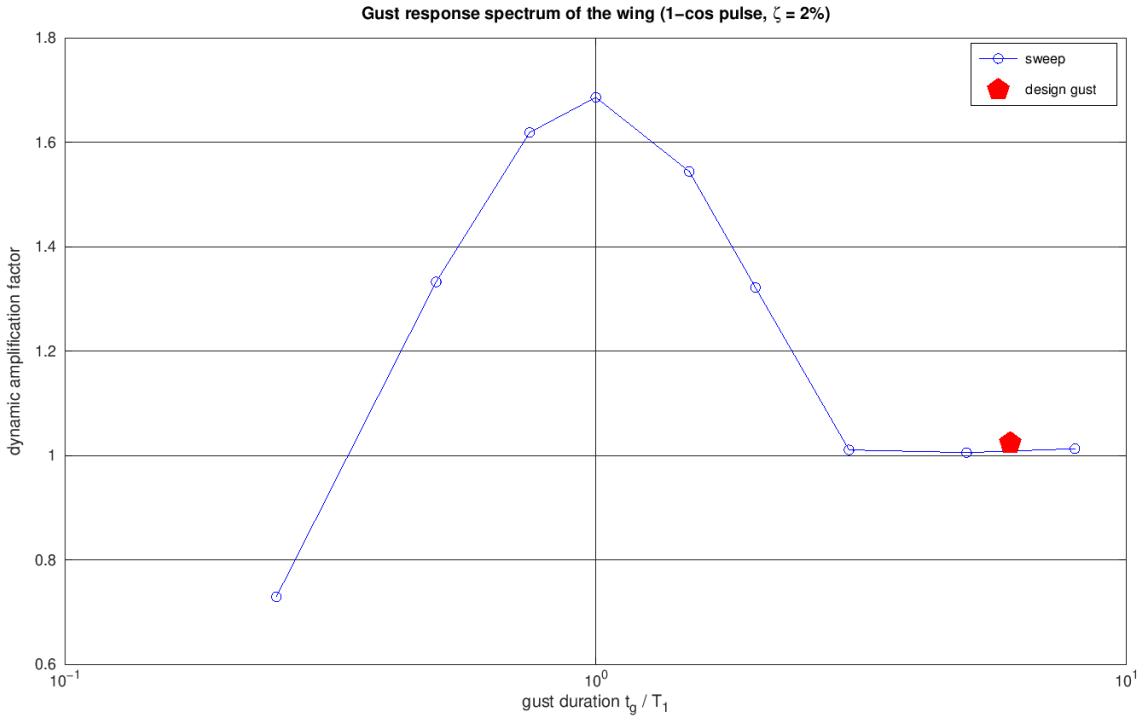


Gust

Gust response spectrum. A sweep of the gust duration shows when dynamics matter:

t_g/T_1	0.25	0.5	0.75	**1.0**	1.5	2	3	5	8
DAF	0.73	1.33	1.62	**1.69**	1.54	1.32	1.01	1.01	1.01

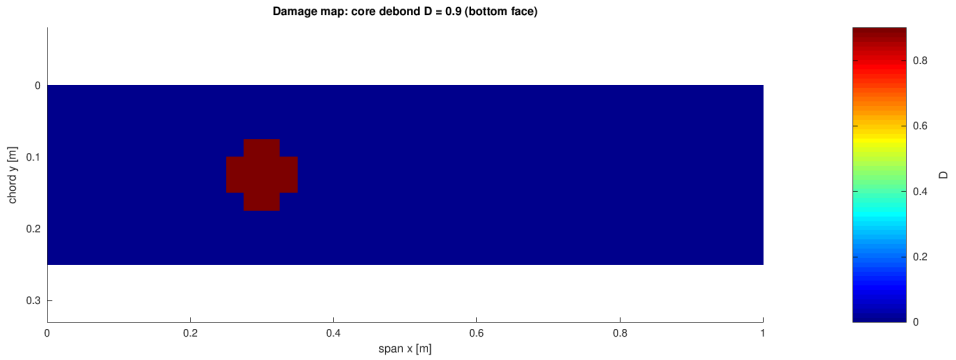
A sharp gust of about 33 ms (one T_1) would amplify the response by 69 %. Damage lowers f_1 and so moves this peak, which is another useful feature for the twist of Milestone 3.

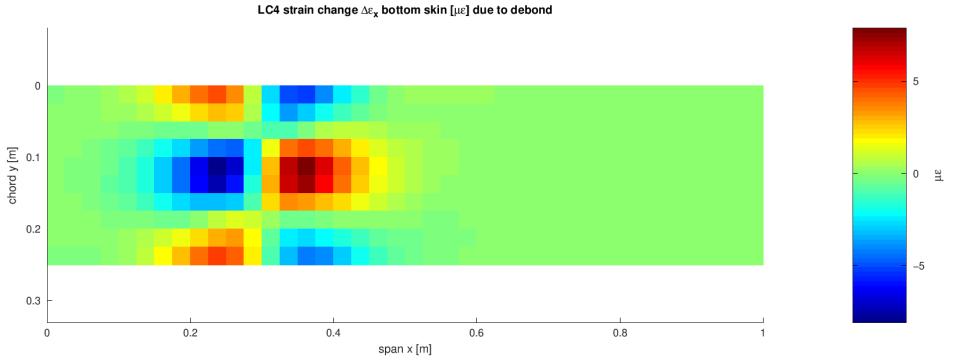


Gust spectrum

5.6 Damage sensitivity — bridge to Milestones 3–5

Damage	D	Elements	Δf/f modes 1… 6 [%]	Δw_tip [%]	min MAC
Core debond (x = 0.3 m)	0.9	12	−0.29, −0.15, **−1.14**, 0.00, −0.34, −0.71	+0.50	0.9993
Skin matrix cracks	0.5	12	−0.07, −0.08, −0.02, −0.02, −0.03, −0.00	+0.13	0.9997
Skin fibre break (near root)	0.5	12	**−0.52**, −0.35, −0.03, −0.09, −0.04, −0.19	+1.05	0.9948
Core crush (x = 0.6 m)	0.8	10	−0.12, −0.55, −0.12, −0.01, −0.61, **−1.37**	+0.27	0.9695





Damage

Key SHM insights, the scientific motivation for the AI part:

1. **Global features are weak.** Frequency shifts are only 0.02 – 1.4 % and MAC stays > 0.97. Temperature or sensor noise of the same order would hide them, so frequencies alone cannot localise damage.
2. **Each damage type has a fingerprint.**
   - A debond hits **2B** most, because 2B carries the highest core shear at x = 0.3 m.
   - A root fibre break hits **1B**, because it has the highest bending strain at the root.
   - A core crush at x = 0.6 m hits **2T**.

   The *pattern* of shifts across modes encodes location and type. This is exactly what a neural network learns well.
3. **Local strain fields are strong.** The debond creates a ±7 με dipole around the damage, about 3–5 % of the local strain. That is easily measurable with FBG sensors. This motivates **physics-informed features** in Milestone 4 (mode-shape curvature, strain-energy damage index) and **full-field reconstruction** in Milestone 6.

5.7 Computational cost (GNU Octave, single core)

Operation	Time
Assembly (400 elements, cached)	0.17 s
6 static load cases	0.01 s
10 modes	0.04 s
Newmark, 2401 steps	4.2 s

So one complete “static + modal” sample costs about 0.25 s, and **10 000 DOE samples take under one hour**. MATLAB is typically faster still.

6. Code map

```
uav_shm_dt/
├── setup_paths.m                  add folders to the path (run first)
├── config/wing_baseline_config.m  Milestone 1: every modelling decision
├── src/materials/   material_library, ply_stiffness, transform_ply,
│                    laminate_section (ABD, As, I0-2), shear_correction
├── src/mesh/        wing_mesh (taper/sweep), rect_mesh, locate_point
├── src/element/     elem_fsdt_q4 (HR / MITC4 / SRI / full), shape_q4, gauss_1d
├── src/assembly/    assemble_system (sparse + cache), apply_bc, dof_index,
│                    wing_layup, build_wing_model
├── src/loads/       lift_distribution (Schrenk + thin airfoil), pressure_load
│                    (singular LE quadrature), point_load, flight_envelope,
│                    gust_load_factor, wing_load_cases
├── src/solvers/     solve_static, solve_modal, solve_newmark, rayleigh_coeffs
├── src/post/        recover_stresses, failure_indices, classify_mode,
│                    plot_field, elem_centroids
├── src/damage/      apply_damage (5 CDM damage modes)
├── src/analytical/  navier_fsdt (static + modal, cross-ply)
├── validation/      val01 ... val09, run_all_validation
├── milestone1/      run_milestone1
├── milestone2/      run_milestone2
├── results/         figures/*.png, data/*.txt|.mat|.md
└── reports/         Milestone1_Report.md, Milestone2_Report.md
```

7. How to verify in MATLAB

```
>> cd uav_shm_dt
>> setup_paths
>> run_all_validation      % expect "42 / 42 required checks passed."
>> run_milestone2          % ~1 min; compare with the tables above
```

These results should agree with Octave to at least 4 significant figures:

Check	Expected
Navier $\bar{w}$ (a/h = 10)	0.662712
FE $\bar{w}$ (a/h = 10, 24×24)	0.662438
Leissa λ_1 (32×32)	3.47174
Wing f_1 … f_3	30.931, 157.294, 162.676 Hz
Wing LC4 tip deflection (mid-chord)	5.892 mm
Wing LC5 Tsai–Wu at x = 50 mm	0.0890
Gust DAF	1.024

The last digits of eigenvalues may differ slightly because MATLAB's `eigs` and Octave's ARPACK differ in tolerance handling. Mode **signs** are normalised, so mode shapes match.

8. Limitations (stated honestly) and what comes next

Limitation	Effect	Plan
Equivalent-single-layer FSDT	No sub-laminate kinematics for large delaminations	Approximated by CDM; layer-wise model possible later
Linear, small deflection	Fine (w_tip/b < 1 %)	—
Quasi-steady aerodynamics	Gust DAF accurate for a clamped wing; no flutter	Unsteady aero if needed
Root fully clamped	Root-corner stress singularity	Design values read at x = 50 mm; spring root in Milestone 6 calibration
Bilinear element	~0.3 % frequency error at 40×10	Enough for the data; Q9 option possible

Next — Milestone 3 (Steps 3–4: Design of Experiments + high-fidelity data generation):

- Latin-hypercube / Sobol sampling over ply angles, thicknesses, materials, loads and damage (type, location, size, severity).
- Batch runs of this solver.
- Outputs per run: frequencies, mode shapes, strains at sensor locations, gust time histories.
- Storage in an HDF5/.mat database ready for feature engineering.

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9. References

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